

1)

To write an Assembly Language Program to disable the keyboard for 5 seconds by masking interrupts and then enable it back.

### PROGRAM:

TITLE KBDMASK.ASM -- Program to disable keyboard for 5 seconds and enable it back

;/\*\*\*\*\*

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;/\*\*\*\*\*

<AMI\_FHDR\_START>

;

; Name: <key.asm>

;

; Description:

<Program to disable keyboard for 5 seconds by masking interrupts and enable it back>

;

<AMI\_FHDR\_END>

\*\*\*\*\*

.model small

.stack

;

data segment

MSG1 DB 13,10,'Keyboard disabled for 5 seconds...\$'

MSG2 DB 13,10,'Keyboard enabled back.\$'

OLDMASK DB 00H

data ends

;-----

code segment

assume cs:code, ds:data

start:

MOV AX, data

MOV DS, AX

;-----

; Read current PIC mask from port 21h

;-----

MOV DX, 21H

IN AL, DX

MOV OLDMASK, AL

;-----

; Mask IRQ1 (keyboard) by setting bit1

;-----

OR AL, 02H

OUT DX, AL

LEA DX, MSG1

MOV AH, 09H

INT 21H

;-----

; Delay for approximately 5 seconds

;-----

CALL DELAY5S

;-----

; Restore original PIC mask

;-----

MOV DX, 21H

MOV AL, OLDMASK

```

    OUT DX, AL
    LEA DX, MSG2
    MOV AH, 09H
    INT 21H
    MOV AH, 4CH
    INT 21H

;-----
; DELAY5S
; Approximate delay routine
;-----

DELAY5S PROC
    PUSH AX
    PUSH BX
    PUSH CX
    PUSH DX
    MOV BX, 5
SEC_LOOP:
    MOV CX, 0FFFFH
OUTER_LOOP:
    MOV DX, 0FFFFH
INNER_LOOP:
    DEC DX
    JNZ INNER_LOOP
    LOOP OUTER_LOOP
    DEC BX
    JNZ SEC_LOOP
    POP DX
    POP CX
    POP BX

```

POP AX

RET

DELAY5S ENDP

code ends

end start

;-----

## OUTPUT:

```
C:\>masm keys.asm;
Microsoft (R) MASM Compatibility Driver
Copyright (C) Microsoft Corp 1993. All rights reserved.

Invoking: ML.EXE /I. /Zm /c /Ta keys.asm

Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: keys.asm

C:\>link keys.obj;

Microsoft (R) Segmented Executable Linker Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp 1984-1992. All rights reserved.

C:\>keys

Keyboard disabled for 5 seconds...
```